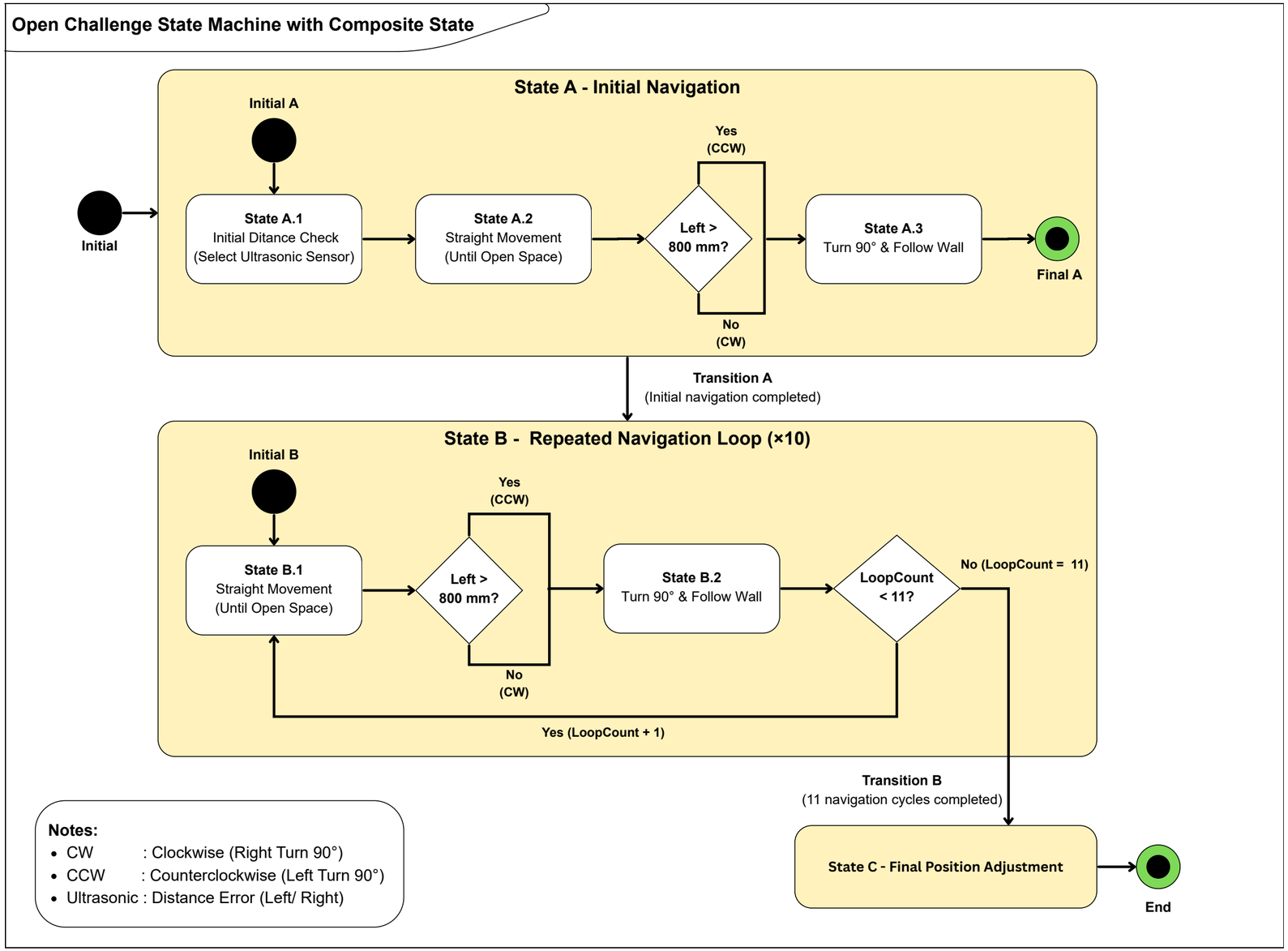

Open Challenge State Machine with Composite State
State A - Initial Navigation
Initial A
Initial
State A.1
Initial Ditance Check
(Select Ultrasonic Sensor)
State A.2
Straight Movement
(Until Open Space)
Yes
(CCW)
Left >
800 mm?
No
(CW)
State A.3
Turn 90° & Follow Wall
Final A
Transition A
(Initial navigation completed)
State B - Repeated Navigation Loop (×10)
Initial B
State B.1
Straight Movement
(Until Open Space)
Yes
(CCW)
Left >
800 mm?
No
(CW)
State B.2
Turn 90° & Follow Wall
LoopCount
< 11?
No (LoopCount = 11)
Yes (LoopCount + 1)
Transition B
(11 navigation cycles completed)
State C - Final Position Adjustment
End
Notes:
CW : Clockwise (Right Turn 90°)
CCW : Counterclockwise (Left Turn 90°)
Ultrasonic : Distance Error (Left/ Right)